

Data Exercise for Candidate Evaluation

January 18, 2026

Instructions

This exercise aims to evaluate your ability to manipulate data to create new variables, provide numerical and visual descriptive statistics, and conduct empirical analysis. This data task contains focuses on one of the topics Pr. Blattman covers in his research: Gang Entry - understanding how kids decide to enter gangs.

If you have access to this pdf, then you must also have been provided with the data to be used for this exercise, the survey instrument (the pdf containing survey questions and possible answers the way they have been presented to surveyees), a variable codebook, and this pdf with instructions. We would like you to complete the tasks presented on this pdf, and **return the following**:

1. A final pdf with your answers to the questions.
2. A clearly organized folder with **replicable** cleaning and analysis codes. (Suggestion for any data task you do: create a dropbox or Github folder for the datatask, within that folder, create additional folders for .do/.R (or other) files, another folder for data, and a final folder for output such as tables and graphs)
3. (Optional) a short feedback on how hard you found each exercise (x/100 and additional comment)

We know there is a lot of text in this pdf, but Pr. Blattman projects rely on a lot of different sources, and we do need to juggle across different files on a daily basis, and in the spirit of guiding you through the different files, we start a lot of these subparts by giving you context or giving you information on where exactly you should find the relevant information. To make things easier, we put in **bold** what the sub-task itself is asking you to do.

1 Gang Entry Task

1.1 Provide basic summary statistics

In this subsection, we would like to draw your attention to understanding the macros of the data. Please **provide summary statistics** on the following:

1. The number of surveys (the N)
2. The survey year (construct it first from instance_time)
3. The nationality of the surveyee
4. The race of the surveyee

If there is missingness in some of the variables, report how many of the observations have this issue. If there are other variables that you find interesting, please do not hesitate to provide summary statistics on them too.

1.2 Creating an important variable: "recruitmentrisk"

Now we would like you to create a variable that we use on a daily basis for our analysis: recruitmentrisk.

1. Open "Data Task/Task 1 documents/Questionnaire_school_survey_2024.docx" and familiarize yourself with "Section 5: Explanation of the 4 Occupational Groups [OCU]" (for context) and "Section 13: Personal Interest in Occupational Groups [EOI]" (for the actual exercise).
 - (a) In short: kids are asked to distribute 10 clones across 4 occupational groups (professional career without finishing high school (G1), career with high school degree (G2), career with higher education degree (G3), and criminal career (G4)). First, they have the choice between all 4 career paths, then we remove the third group (G3 dropped), and then we do remove G2 too. This results in 3 questions for which kids need to report the number of clones they would put in G4.
2. Recruitment risk is measured as the share of clones they put in G4 across the 3 questions. Please **create** this variable and call it recruitmentrisk.
3. **Create a summary table** with main descriptive statistics of the recruitmentrisk variable (mean, min, max, median, quartiles, 90th and 99th percentiles, N).
4. Report as a **histogram or density plot** the distribution of recruitment risk. Add a vertical line for both the mean and the median.

1.3 Creating Priors and Posteriors on Income expectations

Now we would like you to build two new variables: Prior and Posterior Income expectations. Please proceed by doing the following:

1. Open "Data Task/Task 1 documents/Questionnaire_school_survey_2024.docx" and familiarise yourself with the section "10A: Income expectations by occupational group (average person) [EIA]". This question is asked before we run an experiment to evaluate how providing information on average incomes can affect their income expectations.
2. Open "Data Task/Task 1 documents/Baseline Survey Questionnaire (SurveyCTO).xlsx" and familiarize yourself with the corresponding variable names (lines 392 to 433).
3. **Create prior mean income in G1, G2, G3 and G4** variables (one mean per group), and **explain the approach you took to make it**.
4. Redo the same process to **create posterior mean income in G1, G2, G3 and G4** variables ("Section 10B: Income expectations by occupational group (Friends) [EIF]"; lines 586 to 653).

1.4 Recruitment Risk and Mean Income Priors

Now we want to see how recruitment risk correlates with income expectations for each group. Please do the following:

1. **Create a factor variable recruitmentrisk_top_lab** that takes the value of "Low" if the variable recruitmentrisk is in the interval $[0,0.2)$, "Medium" if recruitmentrisk is in the interval $[0.2,0.3)$, and "High" if it is in the interval $[0.3,1]$.
2. Using a for loop, **create 4 density plots** for the distribution of the mean income prior constructed in Section 1.3 by recruitmentrisk_top_lab - one plot per group (G1-G4).
 - (a) To be more precise, you have to create 4 plots, each plot corresponds to a given group in $\{G1, G2, G3, G4\}$, and within each plot, you have 3 overlapping distributions, one distribution for each recruitmentrisk_top_lab. Do all of this using a loop.
 - (b) Bonus points if you put a vertical line for each recruitment risk group's mean.
3. In addition to this, again using a for loop for each group, **create 4 scatter plots** with mean income prior of group g on the Y axis, and recruitmentrisk (continuous, not categorical) on the X axis. **Add a regression line** to each scatter plot.
4. **Report** what you can infer from this exercise.

1.5 Create the Treatment Variable

Now we want you to create a treatment variable for the relevant subset of the surveyed kids:

1. Using the variable `stcs_stroy_lab`, **provide a table reporting the values the variable takes**. What do you notice (if anything)?
2. Now we would like you to create a variable "experiment1" that is defined as `for_mdef_version ≤ 2309151620` (this version of the survey contains the observations that are part of our first experiment). For this subset of observations, report the number of observations, and report a summary table of the values the variable `stcs_stroy_lab` can take.
3. Now create the treatment variable defined as `info_treat == 1` if the person got treated with the information treatment (notice that the video does not count as information treatment since it relates to non-pecuniary outcomes).

1.6 Check Balance

From now on, please focus on the subset of kids that are within the group with `experiment1 == 1` (Ensure you keep the entire dataset as you conduct the analysis on the subgroup - do not drop `experiment == 0` from your dataset). Please **create a balance table** for the demographic variables reported by the surveyees and their recruitment risk. Is there anything to worry about / take into account if you want to conduct an inference on the effect of the information treatment?

1.7 Conduct an analysis of the information treatment experiment

Now that you have all the cards in your hands (demographics, recruitment risk, expected pecuniary reward from being in G1, G2, G3 and G4, `experiment1`, and a treatment variable) we want to see how kids respond to the treatment (which again provided them information on the average income for a person in G1, G2 and G3). Ideally, those who were far away from the actual mean would readjust towards that, and the further they are, the bigger their readjustment, so let's see if that's the case for G3:

1. **Create a new variable** for G3 mean income prior error as the deviation of an individual's mean for G3 earnings prior from 3,000,000 colombian peso (mean earnings in G3 with little experience).
2. **Define each variable** you have created and are planning to potentially use in the analysis (make sure you know exactly what you're dealing with). Furthermore, **report**

the N for each variable, and, if it differs across variables, make sure to explain why it differs and what your preferred approach in dealing with missingness is.

3. Tell us **what analysis you are able to conduct** with these variables, and **what population parameter you are able to estimate**.
4. **Define your regression equation**, and **explain in plain english** why you would or would not include some of the variables you have in your regression.
5. **Conduct the analysis and report your findings. Interpret the coefficients you get.**
6. **Conduct a heterogeneity analysis** by investigating if the treatment effect differs by recruitment risk (do people with higher recruitment risk readjust more, less, or about the same as those with lower recruitment risk?). **Report the results. Interpret the coefficients you get.**